94.3
Pass

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Candidate Information

Email	yourbrajesh@gmail.com
Test	eClerx - Java Coding Test - 3+ Years New
Candidate Packet	View
Taken on	6 Mar 2026 22:45:16 IST
Time taken	34 min 41 sec/ 90 min
Work Experience	3 years
Invited by	Vipul
Invited on	4 Mar 2026 12:57:45 IST

Skill Distribution

No.	Skill	Score
1	Spring Boot Advanced	33%

2	Spring Boot Intermediate	100%	
3	Problem Solving Intermediate	100%	
4	Java Intermediate	100%	

Tags Distribution

Spring Boot	60%	Hard	33%
TDD	0%	Spring	100%
Application Properties Annotations	100%	Medium	100%
Implementation	100%	Arrays	100%
Algorithms	100%	Problem Solving	100%
Greedy	100%	Sorting	100%
Java	100%		

Questions

Spring Boot (Advanced) • 15 / 25

No.	Question	Skill	Score	Code Quality	Time Taken	Status

1	JSON Deserialization Multiple Choice	Spring Boot (Advanced)	0/5	-	15 sec	✗
2	Determine the Output Multiple Choice	Spring Boot (Advanced)	5/5	-	37 sec	✓
3	Response Multiple Choice	Spring Boot (Advanced)	0/5	-	31 sec	✗
4	Configuration Properties Multiple Choice	Spring Boot (Intermediate)	5/5	-	46 sec	✓
5	CSRF Multiple Choice	Spring Boot (Intermediate)	5/5	-	1 min 11 sec	✓

Problem Solving (Intermediate) • 75 / 75

No.	Question	Skill	Score	Code Quality	Time Taken	Status
6	Wind Farms Coding	Problem Solving (Intermediate)	75/75	-	23 min 9 sec	✓

Java (Intermediate) • 75 / 75

No.	Question	Skill	Score	Code Quality	Time Taken	Status

7	Java: DataVille's Optimal Storage Coding	Java (Intermediate)	75/75	-	7 min 36 sec	

1. JSON Deserialization

Incorrect

Multiple Choice | Spring Boot | Hard

Question description

CheckerCreateRequest.java

```
// Annotation here
public class CheckerCreateRequest {
    private String name;
    private String surname;
    public String getName() {
        return name;
    }
    public void setName(String name) {
        this.name = name;
    }
    public String getSurname() {
        return surname;
    }
    public void setSurname(String surname) {
        this.surname = surname;
    }
}
```

CheckerController.java

```
@RestController
@RequestMapping("/checkers")
public class CheckerController {

    @Autowired
    CheckerService checkerService;

    @PostMapping
    public ResponseEntity createChecker(@RequestBody CheckerCreateRequest checkerCreateReque
```

```
st) {
        checkerService.createChecker(checkerCreateRequest);
        return new ResponseEntity(HttpStatus.CREATED);
    }
}
```

Request Json data

```
{
    "checker":{
        "name": "Checker",
        "surname":"Number2"
    }
}
```

A POST request with "Request Json data" is made to "/checkers". What annotation needs to be added to the "CheckerCreateRequest" class so that it can successfully deserialize and process the request?

Interviewer guidelines

@JsonRootName doesn't add checker as a wrapper attribute by itself. To make it successfully deserialize with given json annotations on D should be used together.

Candidate's Solution

Options: (Expected answer indicated with a tick)

- [x] **@JsonRootName("checker")**
- [ ] @JsonName("checker")
- [ ] @JsonTypeName("checker")

@JsonTypeInfo(include = JsonTypeInfo.As.WRAPPER_OBJECT, use = JsonTypeInfo.Id.NAME)

@JsonTypeName("checker")

No comments.

2. Determine the Output

Correct

Multiple Choice | Spring Boot | Hard

Question description

SpringQuestionApplication.java

```
package com.hackerrank.spring;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
@SpringBootApplication
public class SpringQuestionsApplication {
    public static void main(String[] args) {
        SpringApplication.run(SpringQuestionsApplication.class, args);
    }
}
```

Lie.java

```
package com.hackerrank;
import org.springframework.boot.CommandLineRunner;
import org.springframework.stereotype.Component;
@Component
public class Lie implements CommandLineRunner {
    @Override
    public void run(String... args) throws Exception {
        System.out.println("Truth is behind the lies");
        System.out.println("This is working perfectly");
        throw new Exception("Does this work?");
```

```
    }
}
```

Truth.java

```
package com.hackerrank.spring.reality;

import org.springframework.boot.CommandLineRunner;
import org.springframework.stereotype.Component;

@Component
public class Truth implements CommandLineRunner {
    @Override
    public void run(String... args) throws Exception {
        System.out.println("Truth is in your eyes");
    }
}
```

What will be printed on the console when the "SpringBootApplication" class is executed?

Interviewer guidelines

Correct Answer:

Option [A] – "Truth is in your eyes"

Explanation: Spring Boot scans only the package of the `@SpringBootApplication` class and its subpackages. `Truth` is under a subpackage and runs. `Lie` is in a parent package, so it is **not scanned** and its runner does not execute.

Incorrect Answers:

Option [B] – Multiple prints + exceptions

Explanation: `Lie` never runs, so its prints and exception do not occur.

Option [C] – Prints from Lie only

Explanation: `Truth` is discovered and executed; `Lie` is not.

Option [D] – Prints from Lie, then Truth

Explanation: `Lie` is outside the scan path and never executed.

Candidate's Solution

Options: (Expected answer indicated with a tick)

- [x] Truth is in your eyes

- [ ] Truth is in your eyes Truth is behind the lies This is working perfectly And exception occurs and application stops

- [ ] Truth is behind the lies This is working perfectly And exception occurs and application stops

- [ ] Truth is behind the lies This is working perfectly And exception occurs containing “Does this work?” Truth is in your eyes

No comments.

3. Response

Incorrect

Multiple Choice | Hard | Spring Boot | TDD

Question description

Which response will cause the following test case to fail?

```
@Test
  @DisplayName("Filter Price Check 3")
  public void FilterPriceCheck3() throws Exception {
    mockMvc.perform(get("/filter/price/150/500"))
        .andDo(print())
        .andExpect(jsonPath("$[*].product", containsInAnyOrder("Product 1", "Product 2" , "Produc
```

```
t 5")))
    }
```

Candidate's Solution

Options: (Expected answer indicated with a tick)

- [ ] [{"product" : "Product 5"} , {"product" : "Product 1"}, {"product" : "Product 2"}]
- [x] **[{"product" : "Product 5"} , {"product" : "Product 2"}, {"product" : "Product 1"}]**
- [ ] [{"product" : "Product 1"} , {"product" : "Product 2"}, {"product" : "Product 5"}]
- [ ] [{"product" : "product 5"} , {"product" : "product 2"}, {"product" : "product 1"}]

No comments.

4. Configuration Properties

Correct

Multiple Choice | Spring | Spring Boot | Application Properties Annotations | Medium

Question description

An *application.properties* file contains the following properties:

```
app.prop1 = value1
app.prop2 = value2
```

You decided to create a class to hold this configuration. To protect it, the class is defined as final.

```
@Configuration
@ConfigurationProperties(prefix = "app")
public final class AppConfiguration {
  String prop1;
  String prop2;

  public String getProp1() {
    return prop1;
  }

  public void setProp1(String prop1) {
    this.prop1 = prop1;
  }

  public String getProp2() {
    return prop2;
  }

  public void setProp2(String prop2) {
    this.prop2 = prop2;
  }
}
```

What is the result of running the Spring application with this class?

Interviewer guidelines

Spring creates dynamic proxies for classes annotated with `@Configuration` classes. Spring uses CGLIB to extend your class to create proxy. Hence, configuration classes cannot be final. When running the code Spring throws `BeanDefinitionParsingException` and this is the correct option.

Candidate's Solution

Options: (Expected answer indicated with a tick)

- ○ The application runs, the configuration bean is created. and the properties are initialized correctly (prop1=value1, prop2=value2).
- ○ The application runs, and a configuration bean is created, but properties are set to null (prop1=null, prop2=null).
- ● ** A BeanDefinitionParsingException is thrown because the @Configuration bean cannot be marked as final. ** ✓
- ○ An IllegalStateException is thrown because the @Configuration bean cannot have an @ConfigurationProperties annotation.
- ○ A NoSuchMethodException is thrown because the @Configuration bean needs a default constructor.

No comments.

---

5. CSRF

Correct

Multiple Choice | Spring Boot | Medium

Question description

Which Java Spring-based code segment correctly configures CSRF?

Interviewer guidelines

CSRF Protection makes sure that server is not bombarded with unwanted calls. Spring Security enables some features by default like CSRF and Form based authentication.

Candidate's Solution

Options: (Expected answer indicated with a tick)

- ○ private void manage(HttpSecurity http) {   http.csrf().allowed() }
- ○ private void manage(HttpSecurity http) {   http.csrf() }
- ○ private void manage(HttpSecurity http) {   http.csrf().enable() }
- ● ** None of the above. CSRF is configured by default when Spring Security Dependency is added. ** ✓

No comments.

6. Wind Farms

✓ Correct

Coding | Implementation | Medium | Arrays | Algorithms | Problem Solving

Question description

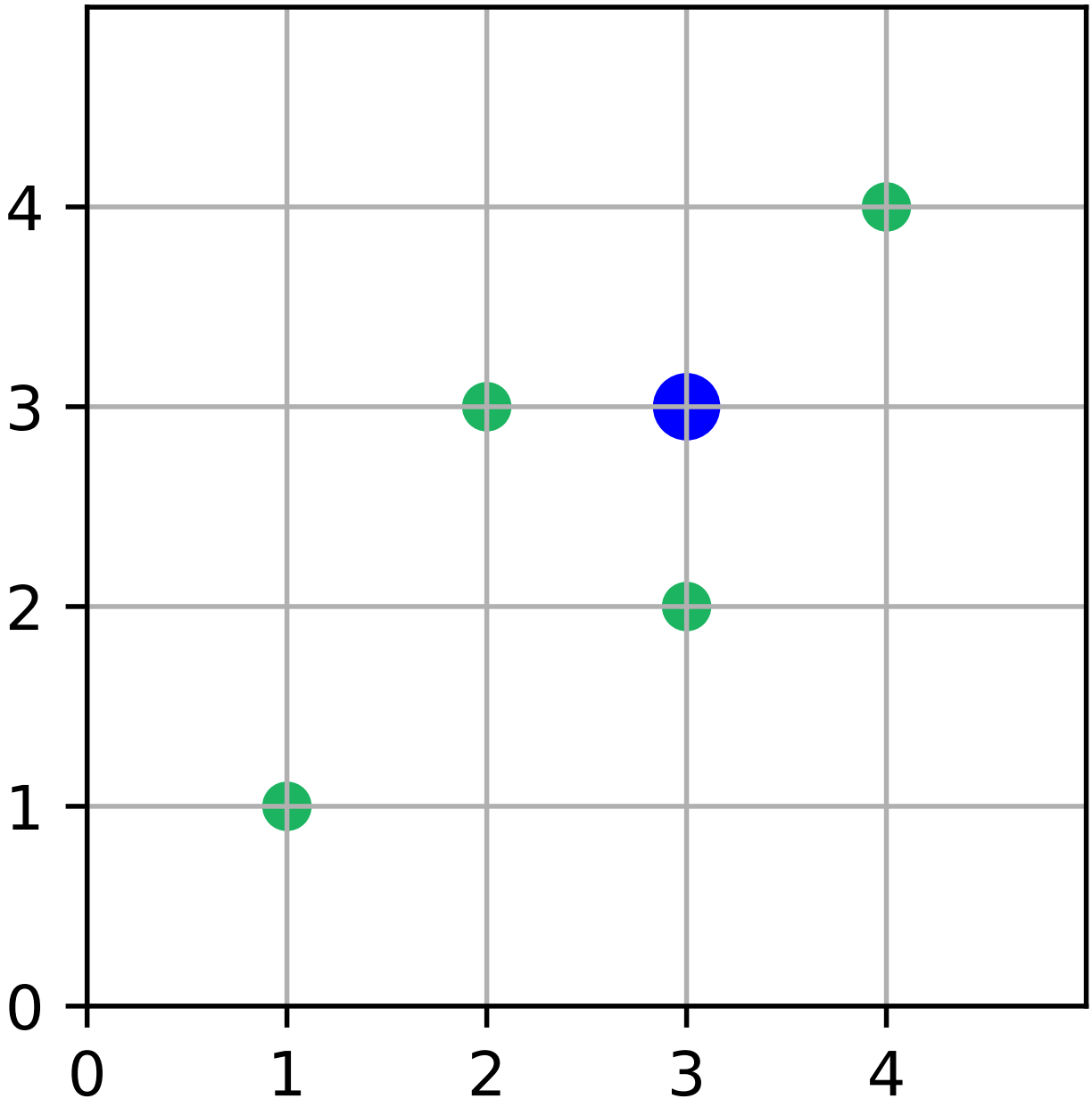
An energy company needs to place a control center to manage a distributed wind farm. Your task is to determine the optimal location for the control center that minimizes the total connection cost.

For each wind farm *i*:

- The farm is located at coordinates *(x[i], y[i])*
- The control center is at coordinates *(cx, cy)*
- The distance is calculated as *|x[i] - cx| + |y[i] - cy|*
- The connection cost is *premium[i] * distance*

Example

x = [1, 3, 2, 4]
y = [1, 2, 3, 4]
premium = [1, 3, 2, 4]



The optimal location is at position [3, 3], resulting in:

Farm #	Location	Distance	Premium	Cost
0	(1, 1)	|1 - 3| + |1 - 3| = 4	1	4 * 1 = 4

1	(3, 2)	|3 - 3| + |2 - 3| = 1	3	1 * 3 = 3
2	(2, 3)	|2 - 3| + |3 - 3| = 1	2	1 * 2 = 2
3	(4, 4)	|4 - 3| + |4 - 3| = 2	4	2 * 4 = 8

The total connection cost is 17, which is the minimum possible.

Function Description

Complete the function *windFarms* in the editor with the following parameter(s):

int premium[n]: the distance multiplier for the cost to connect
int x[n]: *x* coordinates of the wind farms
int y[n]: *y* coordinates of the wind farms

Returns

int: the minimum cost to connect the wind farms

Constraints

- $1 \le n \le 10^3$
- $1 \le x[i], y[i] \le 10^4$
- $1 \le premium[i] \le 50$

▼ INPUT FORMAT FOR CUSTOM TESTING

The first line contains an integer *n*, the size of the array *premium*.
The next *n* lines each contain an element, *premium[i],* where 0 ≤ i < n.
The next line contains an integer *n*, the size of *x[]*.
The next *n* lines each contain an element, *x[i]*.
The next line contains an integer *n*, the size of *y[]*.
The next *n* lines each contain an element, *y[i]*.

Note: *n* is the same value in each instance.

▼ SAMPLE CASE 0

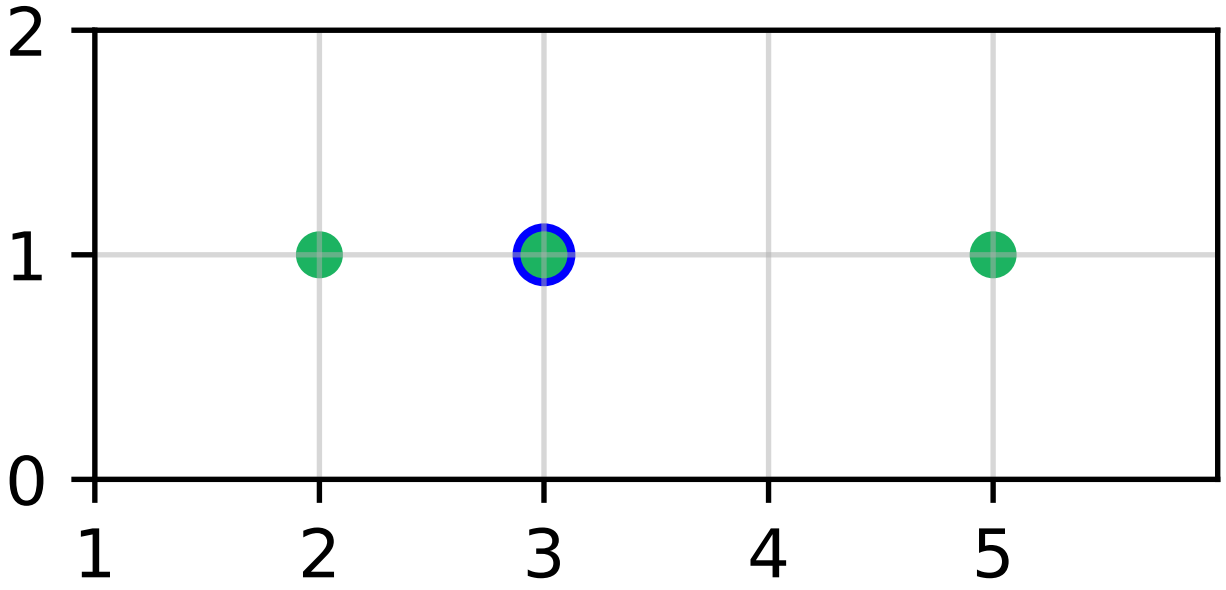
Sample Input 0

```
STDIN    Function
-----    --------
3    →  premium[] size n = 3
1    →  premium = [1, 1, 1]
1
1
3    →  x[] size n = 3
2    →  x = [2, 3, 5]
3
5
3    →  y[] size n = 3
1    →  y = [1, 1, 1]
1
1
```

Sample Output 0

```
3
```

Explanation 0



Locate the controller at farm 1.

Farm #	Location	Distance	Premium	Cost
0	(1, 1)	|1 - 3| + |1 - 1| = 1	1	1 * 1 = 1
1	(3, 1)	|3 - 3| + |1 - 1| = 0	1	0 * 1 = 0

2	(5, 1)	|5 - 3| + |1 - 1| = 2	1	2 * 1 = 2

▼ SAMPLE CASE 1

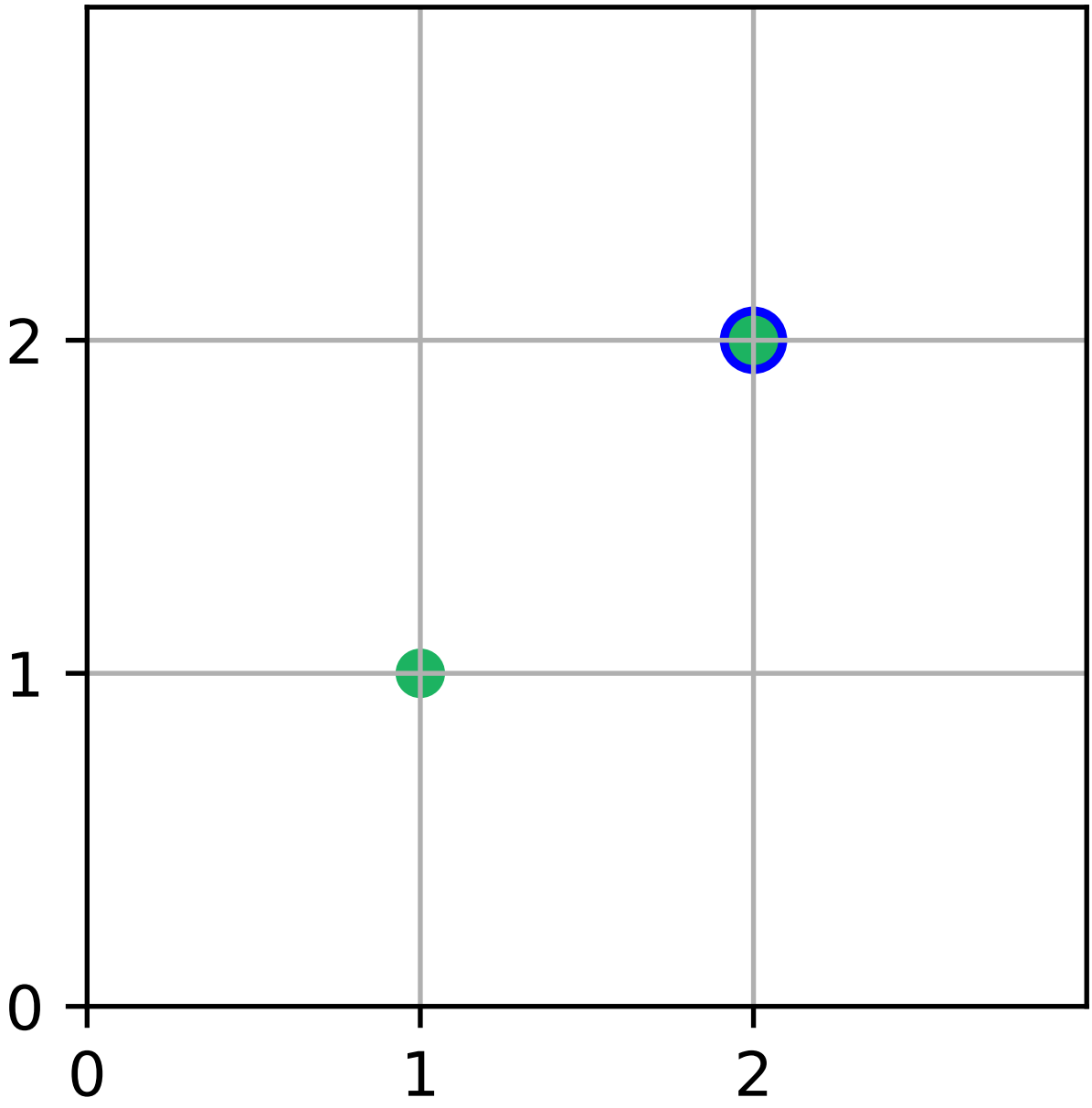
Sample Input 1

```
STDIN    Function
-----    --------
2    →  premium[] size n = 2
1    →  premium = [1, 10]
10
2    →  x[] size n = 2
1    →  x = [1, 2]
2
2    →  y[] size n = 2
1    →  y = [1, 2]
2
```

Sample Output 1

```
2
```

Explanation 1



The sum of the distances to a center at (1, 1), (1, 2), (2, 2), or (2, 1) is 2. Since the premium to connect farm 1 is 10, it is best to locate at position (2, 2) for a cost of 2 * 1 + 0 * 10 = 2.

▼ SAMPLE CASE 2

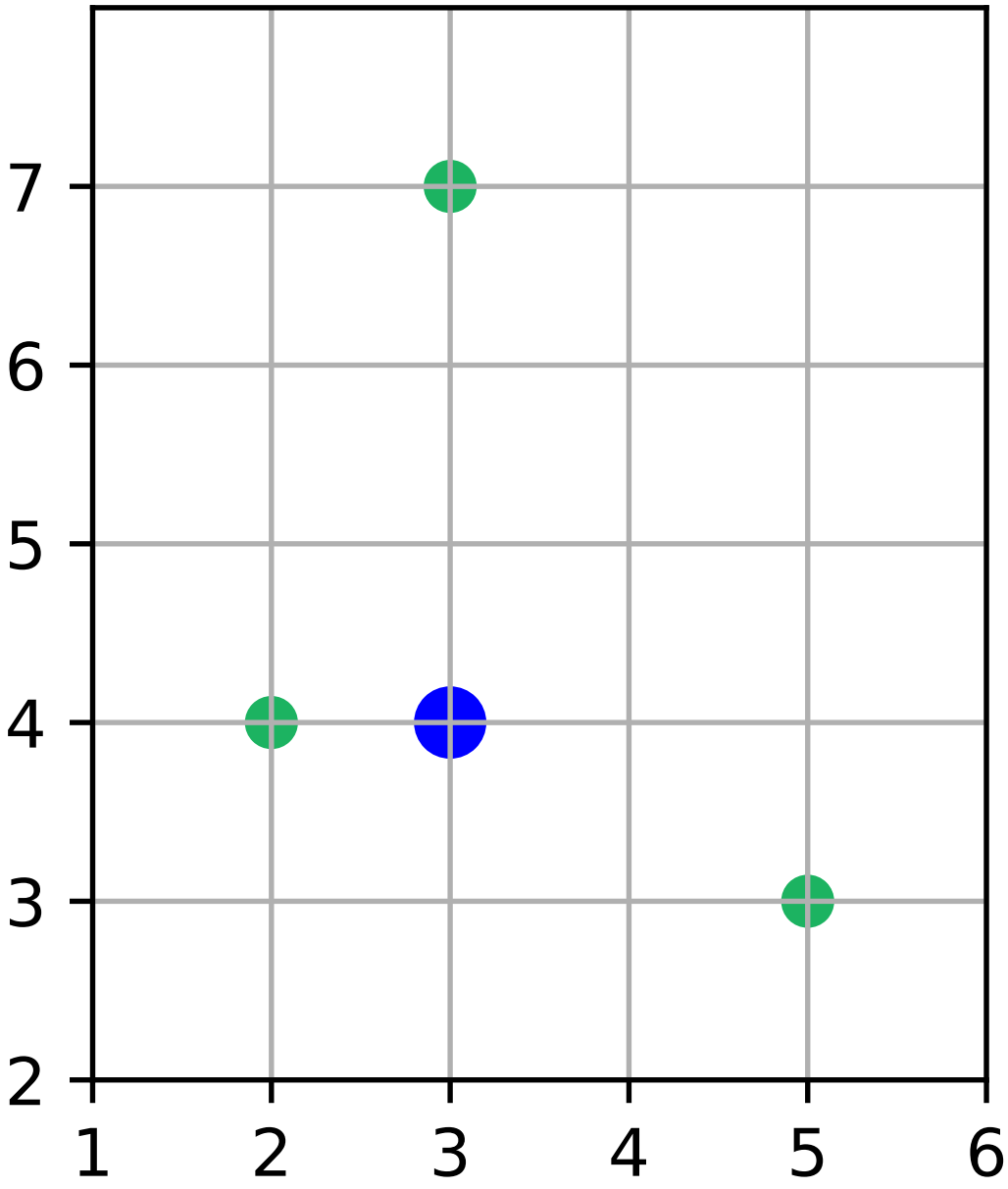
Sample Input 2

```
STDIN    Function
-----    --------
3    →  premium[] size n = 3
1    →  premium = [1, 1, 1]
1
1
3    →  x[] size n = 3
5    →  x = [5, 2, 3]
2
3
3    →  y[] size n = 3
3    →  y = [3, 4, 7]
4
7
```

Sample Output 2

```
7
```

Explanation 2



The optimal location is (3, 4) for a cost of 3 + 1 + 3 = 7.

Interviewer guidelines

▼ SOLUTION

Skills: Loops and counters, I/O

The cost can be calculated for x and y coordinates separately. The function examines each possible x and y coordinate for the control center separately (from 1 to 10000), computes the total cost of connecting all farms for each possible coordinate, and captures the minimum cost found. The ultimate minimum cost is the summation of these minimum costs for x and y coordinates separately.

Optimal Solution

```
def windFarms(premium, x, y):
    minCostX = 10**10
    n = len(x)

    for cx in range(1, 10001):
        cost = 0
        for i in range(n):
            cost += premium[i] * abs(cx - x[i])

        minCostX = min(minCostX, cost)

    minCostY = 10**10
    for cy in range(1, 10001):
        cost = 0
        for i in range(n):
            cost += premium[i] * abs(cy - y[i])

        minCostY = min(minCostY, cost)

    return minCostX + minCostY
```

▼ **COMPLEXITY ANALYSIS**

Time Complexity - O(n)

The universe of x and y values are tested against 10000 values.

Space Complexity - O(1)

Constant space is required regardless of n.

Candidate's Solution

Language used: **Java 21**

```
import java.io.*;
import java.math.*;
import java.security.*;
import java.text.*;
import java.util.*;
import java.util.concurrent.*;
import java.util.function.*;
import java.util.regex.*;
import java.util.stream.*;
import static java.util.stream.Collectors.joining;
import static java.util.stream.Collectors.toList;


import java.io.*;
import java.math.*;
import java.security.*;
import java.text.*;
import java.util.*;
import java.util.concurrent.*;
import java.util.function.*;
import java.util.regex.*;
import java.util.stream.*;

import org.testng.util.RetryAnalyzerCount;

import static java.util.stream.Collectors.joining;
import static java.util.stream.Collectors.toList;



class Result {

    /*
     * Complete the 'windFarms' function below.
     *
     * The function is expected to return an INTEGER.
     * The function accepts following parameters:
     *  1. INTEGER_ARRAY premium
     *  2. INTEGER_ARRAY x
     *  3. INTEGER_ARRAY y
     */

    public static int windFarms(List<Integer> premium, List<Integer> x, List<Integer> y) {
```

```
    // Write your code here
    long minCostX = calculateMinCost(premium, x);
    long minCostY = calculateMinCost(premium, y);
    return (int) (minCostX + minCostY);
    }
    private static long calculateMinCost(List<Integer> premium, List<Integer> coords){
        int n = coords.size();
        List<Coordinate> points = new ArrayList<>();
        for(int i=0; i<n; i++){
            points.add(new Coordinate(coords.get(i), premium.get(i)));
        }
        Collections.sort(points, Comparator.comparingInt(p->p.coord));
        long totalPremium = 0;
        for(int p : premium){
            totalPremium+= p;
        }
        long cumulativePremium =  0;
        int optimalCoord = 0;
        for(Coordinate p: points){
            cumulativePremium+= p.premium;
            if(cumulativePremium>= (totalPremium +1)/2){
                optimalCoord = p.coord;
                break;
            }
        }
        long minCost = 0;
        for(int i=0; i<n; i++){
            minCost+=(long)premium.get(i) * Math.abs(coords.get(i) - optimalCoord);
        }
        return minCost;
    }
    static class Coordinate{
        int coord;
        int premium;
        Coordinate(int coord, int premium){
            this.coord = coord;
            this.premium = premium;
        }
    }

}

public class Solution {
    public static void main(String[] args) throws IOException {
```

```java
        BufferedReader bufferedReader = new BufferedReader(new InputStreamReader(System.in));
        BufferedWriter bufferedWriter = new BufferedWriter(new FileWriter(System.getenv("OUTPUT_PATH")));

        int premiumCount = Integer.parseInt(bufferedReader.readLine().trim());

        List<Integer> premium = IntStream.range(0, premiumCount).mapToObj(i -> {
            try {
                return bufferedReader.readLine().replaceAll("\\s+$", "");
            } catch (IOException ex) {
                throw new RuntimeException(ex);
            }
        })
            .map(String::trim)
            .map(Integer::parseInt)
            .collect(toList());

        int xCount = Integer.parseInt(bufferedReader.readLine().trim());

        List<Integer> x = IntStream.range(0, xCount).mapToObj(i -> {
            try {
                return bufferedReader.readLine().replaceAll("\\s+$", "");
            } catch (IOException ex) {
                throw new RuntimeException(ex);
            }
        })
            .map(String::trim)
            .map(Integer::parseInt)
            .collect(toList());

        int yCount = Integer.parseInt(bufferedReader.readLine().trim());

        List<Integer> y = IntStream.range(0, yCount).mapToObj(i -> {
            try {
                return bufferedReader.readLine().replaceAll("\\s+$", "");
            } catch (IOException ex) {
                throw new RuntimeException(ex);
            }
        })
            .map(String::trim)
            .map(Integer::parseInt)
            .collect(toList());

```

```
        int result = Result.windFarms(premium, x, y);

        bufferedWriter.write(String.valueOf(result));
        bufferedWriter.newLine();

        bufferedReader.close();
        bufferedWriter.close();
    }
}

public class Solution {
    public static void main(String[] args) throws IOException {
        BufferedReader bufferedReader = new BufferedReader(new InputStreamReader(System.in));
        BufferedWriter bufferedWriter = new BufferedWriter(new FileWriter(System.getenv("OUTPUT_PATH")));

        int premiumCount = Integer.parseInt(bufferedReader.readLine().trim());

        List<Integer> premium = IntStream.range(0, premiumCount).mapToObj(i -> {
            try {
                return bufferedReader.readLine().replaceAll("\\s+$", "");
            } catch (IOException ex) {
                throw new RuntimeException(ex);
            }
        })
            .map(String::trim)
            .map(Integer::parseInt)
            .collect(toList());

        int xCount = Integer.parseInt(bufferedReader.readLine().trim());

        List<Integer> x = IntStream.range(0, xCount).mapToObj(i -> {
            try {
                return bufferedReader.readLine().replaceAll("\\s+$", "");
            } catch (IOException ex) {
                throw new RuntimeException(ex);
            }
        })
            .map(String::trim)
            .map(Integer::parseInt)
            .collect(toList());

        int yCount = Integer.parseInt(bufferedReader.readLine().trim());
```

```java

        List<Integer> y = IntStream.range(0, yCount).mapToObj(i -> {
            try {
                return bufferedReader.readLine().replaceAll("\\s+$", "");
            } catch (IOException ex) {
                throw new RuntimeException(ex);
            }
        })
            .map(String::trim)
            .map(Integer::parseInt)
            .collect(toList());

        int result = Result.windFarms(premium, x, y);

        bufferedWriter.write(String.valueOf(result));
        bufferedWriter.newLine();

        bufferedReader.close();
        bufferedWriter.close();
    }
}

```

TESTCASE	DIFFICULTY	TYPE	STATUS	SCORE	TIME TAKEN	MEMORY USED
TestCase 0	Easy	Sample	Success	1	0.0635 sec	37.7 KB
TestCase 1	Easy	Sample	Success	1	0.085 sec	37.8 KB
TestCase 2	Easy	Sample	Success	1	0.0603 sec	37.7 KB
TestCase 3	Easy	Sample	Success	2	0.0581 sec	37.8 KB
TestCase 4	Easy	Hidden	Success	2	0.0547 sec	37.9 KB
TestCase 5	Easy	Hidden	Success	2	0.0696 sec	37.8 KB

TestCase 6	Medium	Sample	Success	6	0.0653 sec	38.2 KB
TestCase 7	Medium	Hidden	Success	6	0.0711 sec	38 KB
TestCase 8	Medium	Hidden	Success	6	0.0802 sec	38.1 KB
TestCase 9	Hard	Hidden	Success	12	0.0904 sec	43.8 KB
TestCase 10	Hard	Hidden	Success	12	0.0942 sec	44.2 KB
TestCase 11	Hard	Hidden	Success	12	0.111 sec	43.6 KB
TestCase 12	Hard	Hidden	Success	12	0.102 sec	44 KB

No comments.

7. Java: DataVille's Optimal Storage

Correct

Coding Greedy Sorting Medium Java

Question description

In the digital city of DataVille, efficient data management is critical. The city's database is represented as a binary array, *databaseStorage*, containing *n* bits of data. Each bit represents one of two types of data:

- Standard data: Represented by 0.
- Priority data: Represented by 1.

To optimize efficiency during peak operational hours, the number of transitions between 0s and 1s in the array must be minimized. Transitions are points in the array where adjacent elements differ (e.g., a 0 followed by a 1 or a 1 followed by a 0).

Operations:

- You can swap any two bits in the array as many times as you like.

The goal is to rearrange the bits to minimize transitions with the minimum number of swaps.

Example

Consider that *databaseStorage* = [1, 0, 1, 0, 1]. One optimal strategy is to swap *databaseStorage[0]* and *databaseStorage[3]*. After performing this operation, the array is [**0**, 0, 1, **1**, 1], which requires only one transition between *databaseStorage[1]* and *databaseStorage[2]*. This requires only 1 swap, so the return value is 1.

Function Description

Complete the function g*etOptimalStorage* in the editor below.

Function Parameters

int databaseStorage[n]: the current organization of the database's storage, with 0s for standard data and 1s for priority data

Returns

int: the minimum number of operations required to minimize transitions between 0s and 1s

Constraints

- $2 \le n \le 10^5$
- *databaseStorage[i]* $\in \{0, 1\}$

▼ INPUT FORMAT FOR CUSTOM TESTING

The first line contains an integer, *n,* the size of *databaseStorage*.
Each of the next lines contains an integer, *databaseStorage[i].*

▼ SAMPLE CASE 0

Sample Input 0

```
STDIN       FUNCTION
-----       --------
```

```
7          databaseStorage[] size n = 7
1          databaseStorage = [1, 0, 1, 1, 0, 0, 1]
0
1
1
0
0
1
```

Sample Output 0

```
1
```

Explanation

One optimal strategy is to swap *databaseStorage[1]* (which is 0) with *databaseStorage[6]* (which is 1). After performing this operation, the array becomes [1, **1**, 1, 1, 0, 0, **0**]. This solution requires only 1 swap, so the output is 1.

▼ SAMPLE CASE 1

Sample Input 1

```
STDIN     FUNCTION
-----     --------
9         databaseStorage[] size n = 9
1         databaseStorage = [1, 0, 0, 1, 1, 0, 0, 0, 1]
0
0
1
1
0
0
0
1
```

Sample Output 1

```
2
```

Explanation

One optimal strategy of operations is:

1. Swap *databaseStorage*[1] and *databaseStorage*[8]. After performing this operation, the array becomes [1, **1**, 0, 1, 1, 0, 0, 0, **0**].

2. Swap *databaseStorage*[2] and *databaseStorage*[4]. After performing this operation, the array becomes [1, 1, **1**, 1, **0**, 0, 0, 0, 0].

Now there is only one transition. The minimum number of swaps is 2.

Interviewer guidelines

▼ SOLUTION

The solution is to count the number of 0s and 1s in the array. Then, we count the number of 0s and 1s that are out of place. The answer is the minimum of the two counts.

```
class Result {

    /*
     * Complete the 'getOptimalStorage' function below.
     *
     * The function is expected to return an INTEGER.
     * The function accepts INTEGER_ARRAY databaseStorage as parameter.
     */

    public static int getOptimalStorage(List<Integer> databaseStorage) {
        int zeros = 0, ones = 0;

        for (int brick : databaseStorage) {
            if (brick == 0) {
                zeros++;
            } else {
                ones++;
            }
        }

        int startZero = 0, startOne = 0;

        for (int i = 0; i < ones; ++i) {
            if (databaseStorage.get(i) != 1) {
                startOne++;
            }
        }

        for (int i = 0; i < zeros; ++i) {
            if (databaseStorage.get(i) != 0) {
                startZero++;
            }
```

```
        }

        return Math.min(startZero, startOne);
    }
}
```

▼ COMPLEXITY

Time Complexity - O(n)

To traverse over the array, we use a loop that runs n times. The complexity is O(n).

Space Complexity - O(1)

We use only 2 variables, hence the space complexity is O(1).

Candidate's Solution

Language used: **Java 21**

```java
import java.io.*;
import java.math.*;
import java.security.*;
import java.text.*;
import java.util.*;
import java.util.concurrent.*;
import java.util.function.*;
import java.util.regex.*;
import java.util.stream.*;
import static java.util.stream.Collectors.joining;
import static java.util.stream.Collectors.toList;


class Result {

    /*
     * Complete the 'getOptimalStorage' function below.
     *
     * The function is expected to return an INTEGER.
     * The function accepts INTEGER_ARRAY databaseStorage as parameter.
     */

    public static int getOptimalStorage(List<Integer> databaseStorage) {
        int count0 = 0;
```

```
        int count1 = 0;
        int n = databaseStorage.size();

        for(int bit : databaseStorage){
            if(bit == 0){
                count0++;
            }else{
                count1++;
            }
        }
        int swaps1 = 0;
        for(int i =0; i< count0; i++){
            if(databaseStorage.get(i) == 1){
                swaps1++;
            }
        }
        int swaps2 = 0;
        for(int i = 0; i< count1; i++){
            if(databaseStorage.get(i) == 0){
                swaps2++;
            }
        }
        return Math.min(swaps1, swaps2);

    }

}

public class Solution {
    public static void main(String[] args) throws IOException {
        BufferedReader bufferedReader = new BufferedReader(new InputStreamReader(System.in));
        BufferedWriter bufferedWriter = new BufferedWriter(new FileWriter(System.getenv("OUTPUT_PATH")));

        int databaseStorageCount = Integer.parseInt(bufferedReader.readLine().trim());

        List<Integer> databaseStorage = IntStream.range(0, databaseStorageCount).mapToObj(i -> {
            try {
                return bufferedReader.readLine().replaceAll("\\s+$", "");
            } catch (IOException ex) {
                throw new RuntimeException(ex);
            }
        })
```

```
            .map(String::trim)
            .map(Integer::parseInt)
            .collect(toList());

        int result = Result.getOptimalStorage(databaseStorage);

        bufferedWriter.write(String.valueOf(result));
        bufferedWriter.newLine();

        bufferedReader.close();
        bufferedWriter.close();
    }
}

```

TESTCASE	DIFFICULTY	TYPE	STATUS	SCORE	TIME TAKEN	MEMORY USED
TestCase 0	Easy	Sample	Success	1	0.0517 sec	37.7 KB
TestCase 1	Easy	Sample	Success	1	0.0619 sec	37.6 KB
TestCase 2	Easy	Sample	Success	1	0.0489 sec	37.4 KB
TestCase 3	Easy	Hidden	Success	3	0.0528 sec	37.6 KB
TestCase 4	Easy	Hidden	Success	3	0.06 sec	37 KB
TestCase 5	Easy	Hidden	Success	3	0.0543 sec	37.9 KB
TestCase 6	Easy	Hidden	Success	4	0.0568 sec	37.5 KB
TestCase 7	Easy	Hidden	Success	4	0.0509 sec	37.4 KB

TestCase 8	Easy	Hidden	Success	5	0.0561 sec	37.5 KB
TestCase 9	Easy	Hidden	Success	5	0.0637 sec	37.5 KB
TestCase 10	Easy	Hidden	Success	5	0.0481 sec	37.8 KB
TestCase 11	Medium	Hidden	Success	7	0.0652 sec	39.8 KB
TestCase 12	Medium	Hidden	Success	8	0.0733 sec	39.5 KB
TestCase 13	Hard	Hidden	Success	10	0.0711 sec	39.5 KB
TestCase 14	Hard	Hidden	Success	15	0.2183 sec	69.2 KB

No comments.